

FLATTRIKE · DESIGN DECISIONS REGISTER

FlatTrike design decisions register

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DOCUMENT CONTROL

Document ID	FTK-DEC-001	Version	0.5
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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/flattrike		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-10-03	Amish Chadha	Fork direction decided by Amish 2026-10-03 (knuckle forks turned to trail, 35 mm mechanical trail); knuckle fork added to the items to confirm; value engineering restated (BOM line 5 \$100; FTK-CAL-001 v0.7)
0.4	2026-10-02	Amish Chadha	Value engineering restated after the 2026-10-02 decisions were carried into the design (lock stops in BOM line 3; FTK-CAL-001 v0.6)
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for all open decisions 1 to 10 on 2026-10-02 (FTK-DDR-003 accepted; rating 148 and 138 kg, R8 72 kg, R14 60 min for knuckle post plates; steering stops at the knuckle posts); moved to decisions made
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items from the review note, the decision records and the build work

FlatTrike design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

Table 1. Items to confirm when parts are bought.

#	What to confirm	Why it matters	Source
1a	The two knuckle forks: 20 in steel, 100 mm dropouts, threaded 1 1/8 in steerer, straight legs parallel to the steerer with 30 to 40 mm offset (measure it); 22 mm legs	Fitted turned round, the offset is the trail (30 to 40 mm); a bent-leg fork would put the steering arm clamp on the bend	FTK-CAL-001 section 5a; decided 2026-10-03
1	The steering column fork: threaded 1 1/8 in steel fork with a separate forged crown (roadster type) solid enough to drill two 9 mm holes 32 mm either side of the steerer; steerer at least 250 mm	The drop arm bolts under the crown	FTK-DDR-003, P8
2	A quill stem with about 350 mm of quill for a 1 1/8 in threaded steerer, 70 mm or more inside the steerer	It sets the handlebar height of 850 mm	FTK-DDR-003, P8
3	Front fork leg diameter (22 mm assumed) and M6 U-bolts that fit it; the U-bolts' grip holds the steering arm under a firm push on the bar	The steering arms are held by friction on the leg	FTK-DDR-003, P10
4	Front drum hub body 90 mm across or less where the steering arm clamp plate passes over it	The clamp plates clear a 90 mm hub by about 6 mm	FTK-DDR-003, P10
5	Rod ends for M8 bolts with at least $\pm 10^\circ$ misalignment	The drag link slopes about 6°	FTK-DDR-003, P9
6	Headset cups' spigots long enough to pass a 3 mm plate and still seat 10 mm in the head tube	The cups clamp the yokes and collar plates	FTK-DDR-003, P7 and P11
7	A cartridge bottom bracket for a 68 mm shell that works with 66 mm between the cup flanges (60 mm shell and two 3 mm plates)	The cups clamp the spine plates	FTK-DDR-003, P5
8	Hub axles long enough for 3 mm dropouts plus non-turn washers and nuts; front drum reaction arms that fit the forks	The dropouts are thinner than a bicycle frame's	FTK-DDR-003, P3
9	Drum brake torque of about 37 N·m per wheel, and the 3-speed hub's input torque limit for 32/24 gearing	R11 and the hub's life	FTK-CAL-001; BOM lines 6 and 7
10	The laser or waterjet shop holds slot widths to about 0.1 mm (3.4 mm slots for 3 mm tabs)	Tabs must enter by hand and not rattle	FTK-DDR-003, P1
11	Seat tube 31.8 x 2.3 mm whose bore takes a 27.2 mm seatpost	The seatpost clamps in it	FTK-DDR-003, P6

■ Value engineering

Value-engineering target: USD 800 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 736 (USD 64 under the target). Main cost drivers and savings worth trying:

- The largest prototype lines are the two front wheels with drum brakes (USD 110), the rear wheel with 3-speed drum hub (USD 95), the steering column, knuckles and linkage (USD 100) and the front bed and knuckle post plates with the steering lock stops (USD 89). The construction changes (FTK-DDR-003) added USD 15 (USD 709 to USD 724), the two lock stops and their bolts USD 4 (USD 724 to USD 728), and the straight-leg knuckle forks fitted turned to trail USD 8 (USD 728 to USD 736).

- The optional route C assist kit (USD 300, without the SwapCell pack) is outside the prototype target.
- Production target: USD 300 per trike at 100 units (R18) against an estimate of about USD 471, USD 171 over the target. Bought bicycle parts are USD 329 of it at 65 % of retail, with USD 74 of steel (one full sheet), USD 17 of cutting, the plywood box at 60 % and USD 15 of assembly labour.
- Savings worth trying: wholesale prices for wheels, hubs and forks once the first partner supplies them (FTK-DDR-002, item 18), and the one-sheet nest that already keeps the steel and cutting low.

■ Decisions made

Table 2. Decisions made.

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 9, 11 and 12: tadpole front loader; box steering first with Ackermann assessed; no assist in the prototype, route C (SwapCell) preferred; 3 mm S235 or A36 plate, painted; M8 8.8 bolts with all-metal locknuts; bought head tube and bottom bracket sections; 20 in wheels with drum brakes and a 3-speed rear hub; design changes before relaxing R8 and R10; \$300 production target (R18); budget_usd \$800 for the pedal-only prototype; problem line reworded	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	FTK-DDR-001
2026-09-25	Cross-cutting: SwapCell interface v0.3 items W, C and V for assist route C; a SwapCell pack is priced once in the SwapCell repo; partners are picked per area later	Amish, same instruction	FTK-DDR-001
2026-09-25	Items 13 to 19: Ackermann steering with a fixed box; R8 70 kg; R10 rider-only 0.22 g with a cornering-speed label; 140 kg cargo for riders over 80 kg; R14 90 min for a spine side plate; keep the \$300 target; 200 mm head tube	Amish: "i accept all your recommendations, go with them across all repos."	FTK-DDR-002
2026-09-25	TRL 4 on hold; the repo stays at TRL 3	Amish's instruction	project.yaml ; review note
2026-09-26	FlatTrike in the first batch of product renders	Amish	Review note, 2026-09-26
2026-09-30	Write the illustrated build plan and make the design physically buildable as it is drawn; keep open decisions out of the build plan, in this register	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." and "don't log outstanding decisions in this build plan - that is not the place for it. that should be in a separate design document logged and named as such"	FTK-DDR-003 (Draft, open for review); this register
2026-10-02	Design for construction accepted: the changes P1 to P15 of FTK-DDR-003 and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	FTK-DDR-003, Tables 1 and 2
2026-10-02	Rated load (option c): rated now at 148 kg of cargo with a rider up to 80 kg, or 138 kg with a rider up to 90 kg, and R8 relaxed to 72 kg; the lightening windows are tried in the FEA session, and the final load label is set from the weighed trike	Amish: "i approve your recommendations for all 555 open decisions."	FTK-DDR-003, A1
2026-10-02	Repair time: 60 min allowed for a knuckle post plate (R14)	Amish: "i approve your recommendations for all 555 open decisions."	FTK-DDR-003, A2

Date	Decision	Decided by	Record
2026-10-02	Assembly time: about 4.04 h accepted against R7's 4 h, timed at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	FTK-DDR-003, A3
2026-10-02	Steering lock stops (option b, changed from a): a stop at each knuckle post that its steering arm clamp plate meets at 40 degrees, fitted before the first ride	Amish: "i approve your recommendations for all 555 open decisions."	Build plan work; this register
2026-10-02	Steering ratio: the 1:1 parallelogram linkage is kept	Amish: "i approve your recommendations for all 555 open decisions."	FTK-DDR-003, P9
2026-10-02	Steering geometry: caster, trail, kingpin inclination, scrub radius and self-centring are studied on paper in the FEA session, with the bought fork's rake as input, before the first ride	Amish: "i approve your recommendations for all 555 open decisions."	FTK-PRC-001, open questions
2026-10-02	First partner and city: kept open under the portfolio rule, chosen as a city with dense goods delivery on narrow streets, a laser-cutting shop and a bicycle parts market. First candidate type to approach: a cycle-rickshaw or cargo-bike programme such as those the Institute for Transportation and Development Policy (ITDP) has documented in South Asia	Amish: "i approve your recommendations for all 555 open decisions."	FTK-DDR-002, item 10
2026-10-02	Render frame colour: teal and graphite for the renders only; the coating is chosen with the partner	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, item 1
2026-10-02	Appearance model details (bent fork legs, round tyre section, box hardware positions, flat-cut chain guard, brake cable runs) accepted as appearance only	Amish: "i approve your recommendations for all 555 open decisions."	Review note 2026-09-26, items 4 to 8
2026-10-03	Fork direction (steering geometry before the first ride): each knuckle fork is a straight-leg fork with 30 to 40 mm offset, fitted turned round so the wheel trails its vertical steering axis; 35 mm of mechanical trail, no caster; bed rails notched and knuckle post lip trimmed for the swinging tyre and crown; wheelbase 1.415 m, turning circle 5.52 m	Amish: "FlatTrike - i accept your design recommendation, proceed and execute the solution for fork direction or caster"	FTK-CAL-001 section 5a; review note 2026-10-03